ArtBench-10 Generative Modeling: A Comparative Study of VAE, GAN, and Diffusion Models

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Abstract. Nothing yet.

1 Introduction

2 Problem Formulation

3 Methodological Framework

3.1 Data Processing Pipeline

VAE

DCGAN

Diffusion Models

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3.3 Theoretical Rationale for Hyperparameter Choices

Variational Autoencoders (VAE)

Generative Adversarial Networks (GAN)

Diffusion Models

4 Experimental Design

4.1 Hyperparameter Optimization Strategy

We performed hyperparameter optimization with Bayesian optimization (Optuna). To keep the search feasible on standard hardware, tuning was conducted on a development subset comprising 20% of the training data. For each model

family, candidate configurations were trained on a training split and selected using performance on a held-out validation split. We minimized the following surrogate objectives: (i) *VAE*: validation reconstruction loss; (ii) *GAN*-based models: validation sum of discriminator and generator losses; (iii) *Diffusion models*: validation denoising loss under a fixed Gaussian noise schedule. The hyperparameter search spaces are summarized in Table 1, and the best configurations found on the studies are reported in Table 2. We used validation losses as surrogates during tuning because FID/KID are computationally expensive and high-variance at the per-trial scale: they require generating large sample sets and repeated feature extraction. In contrast, validation losses are available at each epoch and exhibit lower variance which improves sample efficiency under limited computational resources. We note, however, that GAN losses are far from beeing perfect proxies for perceptual sample quality; therefore, we use this objective only for hyperparameter selection during tuning and rely on qualitative inspection for the selected final parameters.

Table 1. Hyperparameter search spaces used in Bayesian optimization.

Model	**Parameters**	**Values / Range**
VAE	latent_dim, base_channels, batch_size	$\{16, 32, 64\}$; $\{16, 32, 64\}$; $\{32, 64, 128\}$
	lr, β	$[10^{-4}, 3 \times 10^{-3}]$ log-uniform; $[10^{-6}, 10^{-3}]$ log-uniform
DCGAN	latent_dim, feature_maps, batch_size	$\{128, 256, 512\}$; $\{32, 64\}$; $\{32, 64, 128\}$
	lr, β_1, real_label	$[10^{-4}, 5 \times 10^{-4}]$ log-uniform; $[0.3, 0.7]$; $[0.85, 1.0]$
PixelUNet	model_channels, batch_size, num_timesteps	$\{32, 64, 96\}$; $\{16, 32, 64\}$; $\{500, 1000\}$
	lr	$[10^{-4}, 5 \times 10^{-4}]$ log-uniform
LatentDenoiser	model_channels, num_res_blocks, batch_size	$\{32, 64, 96\}$; $\{2, 3, 4\}$; $\{16, 32, 64\}$
	lr, num_timesteps	$[10^{-4}, 5 \times 10^{-4}]$ log-uniform; $\{500, 1000\}$

Table 2. Best configurations obtained.

Parameter	VAE	DCGAN	PixelUnet	LatentDenoiser
Best objective value	0.02658	1.68158	0.0358	0.2598
Learning rate	0.0005317	0.0001261	0.0004748	0.0001769
Batch size	32	32	16	32
Latent dimension	32	256	64	64
Base channels / feature maps	64	64	64	64
β (VAE)	1.14×10^{-6}	–	–	–
β_1 (Adam, DCGAN)	–	0.4813	–	–
real_label (DCGAN)	–	0.9836	–	–
num_timesteps (diffusion)	–	–	1000	1000
num_res_blocks (LatentDenoiser)	–	–	–	4

4.2 Evaluation Protocol

Across the selected configurations (Table 2), the optimization process consistently favored smaller batch sizes. This aligns with the well-documented generalization gap observed in large-batch training [2]. Specifically, the inherent noise in small-batch stochastic gradients serves as a form of implicit regularization, facilitating the escape from sharp local minima and leading the model toward flatter, more robust regions of the loss landscape. Regarding the learning rates, the observed values are relatively low, consistent with the need for stable convergence and stability during training, especially for GANs which are known to be sensitive to hyperparameter settings. Latent dimensions reveal important differences across model families: the VAE favored a much smaller latent space (32) compared to the DCGAN (256), which may reflect the different in the model's architectures: GANs don't pay an explicity penalty (KL divergence) for using larger latent spaces, while VAEs are encouraged to find compact representations. A larger latent space may encourage more diverse samples as long as the generator actually uses it. For the VAE, the optimal β value is very low, indicating that the model prioritizes reconstruction fidelity over latent space regularization. For the DCGAN, the optimal β_1 value is around 0.48, which is close to the commonly used value of 0.5 [5], suggesting a balance between stability and convergence speed. The real_label parameter being close to 1 (0.9836) indicates that the discriminator benefits from a small anti-overconfidence margin, which can help prevent it from becoming too confident and overpowering the generator during training without weakening the learning signal too much.

5 Results and Interpretation

5.1 Ablation Studies and Discussion

6 Conclusion

Use of External Resources and AI Tools

This project utilized ArtBench-10 (Kaggle). Gemini assisted in LaTeX formatting and analysis verification.

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